Koç University
College of Engineering
ELEC 491 - Design Project Proposal

Spring 2025

Interactive Ultrasonic Theremin

Participant Info:

Name	ID	Email	Phone
1 Yaşar Aykut Altuntop	75992	yaltuntop20@ku.edu.tr	+90 538 811 0730
2			
3			
4			
5			

Project Supervisor:

Name
Levent Beker

Name of TA:

Name
BITOP MAITRA

Abstract

Theremin is a contactless electronic musical instrument that usually consists of two metal antennas working basically as capacitor plates to create a varying capacity with users' hands. Generally, the right hand of the performer controls the pitch, and the left-hand controls the volume, although in some versions volume is controlled by a knob and there is only a pitch antenna.

The Interactive Ultrasonic Theremin aims to design a musical instrument that uses ultrasonic sensors to detect hand movements and generate musical notes based on the distance, taking a different approach from a traditional theremin which uses capacitance variations, but still with similar functionality. The pitch and volume of the sound will be determined by the distance and position of the performer's hands relative to the sensors, allowing for a gesture-based musical experience.

The project will consist of two axes of ultrasonic sensors to determine the distance of the user's hands, It will be used with either one or two hands to control pitch and volume based on the choice of design implementation. A microcontroller will process these data and decide on the pitch and the volume. After this stage the corresponding sound will be generated using a DAC-amplifier-speaker system or by a PC software to enrich the musical options.

This project is expected to be completed within the semester and will be presented during the senior design day. A detailed project report documenting the design process, implementation, and results will be submitted at the end of the semester.

Table of contents

Introduction 4
Concept 4
Objectives 4
Background 4
Background and Significance 4
Current State of the Art 5
Problem to be Addressed 5
S/T methodology and associated work plan 5
Methodology 5
Work Package Descriptions 6
Demonstration 10
Impact 10
Risk analysis 10
Gantt Chart 11
Ethical Issues 11
References 11

Introduction

Concept

The **Interactive Ultrasonic Theremin** device is a musical instrument that has similar functionality to the traditional theremin, but unlike the capacitance-based traditional theremin, it uses **ultrasonic sensors** to change pitch and volume.

This device will basically work with the concept of changing **pitch** and **volume** by measuring distance with **ultrasonic sensors** positioned in 2 different directions. The MCU to be used will process the sensor data in real-time and provide a sound output at the specified pitch and volume. The method of producing the specified sound output will be decided during the design process, the processed signal will be amplified and filtered and given to a speaker, and a smooth and clear sound will be provided.

Since **traditional theremins** are **capacitance-based**, they are affected by environmental factors, this device that will work with ultrasonic sensors will prevent this and will also add a different approach to musical instruments. At Senior Design Day, a product that can be experienced by users completely like an instrument will be built and every stage of production will be reported and collected in the final project report. The technical evaluation of the device will be based on user experience factors such as sensor accuracy, total latency of the system, sound quality, smoothness of sound transitions, and playability.

Objectives

The Interactive Ultrasonic Theremin project aims to develop a functional, gesture-controlled musical instrument that replicates the playing experience of a traditional theremin using ultrasonic sensors. By the end of the project, the following outcomes are expected:

Fully Functional Prototype: At the end of the project, the theremin-like device will be fully functional and can be played by users within their abilities.

Accurate Hand Tracking System: The device aims to track hand movements using ultrasonic sensors and accurately determine the corresponding pitch and volume.

Reliable Sound Generation Method: The sound generation method has not been decided yet but it could involve a Digital-to-Analog Converter (DAC), Pulse-Width Modulation (PWM), or another technique, depending on the outcomes of the design phase. Nevertheless, the sound generation outcome needs to be a clear sound in accordance with the desired pitch and volume with the least possible delay.

User Interaction and Playability: Real-time responsiveness is also important as the device aims to be in a final state so that all users are able to play it.

Documentation and Performance Evaluation: The final project report is going to be prepared so that It includes all the mechanical and electronic design steps and the used firmware with detailed explanations.

By the end of the semester, the project should deliver a working and well-documented interactive ultrasonic theremin, demonstrating an alternative approach to musical expression using sensor technology. The success of the project will be measured based on the system's ability to accurately track hand movements, generate smooth and expressive sound, and provide a satisfying user experience.

Background

Background and Significance

Theremin is a musical instrument that can generate sounds of different frequencies and volumes by capacitance changes that occur as a result of the movement of the hands of the users without any contact with the device. It was invented by Leon Theremin in 1920. He first named this device "Aetherphone" because it created music

from ether, which was thought to fill space at the time and provide a medium for light to spread[1].

While traditional versions work on the heterodyne oscillator principle, modern versions provide more advanced music support by supporting digital processing or MIDI. However, this instrument, which works with capacitive sensors, can be quite sensitive to environmental conditions.

Current State of the Art

Traditional Theremin Technology

- **Moog Etherwave Theremin** – A modern analog theremin that is said to be "An authentic adaptation of Bob Moog's theremin designs"[2].

Ultrasonic Gesture-Based Instruments

- **Ultrasonic Theremin Demonstration:** https://vimeo.com/groups/arduinoprojects/videos/88496135

Problem to be Addressed

This project explores an **alternative approach** to theremin design by replacing capacitive sensing with **ultrasonic distance measurement**.

The key challenges include achieving smooth frequency and volume control to make it playable, ensuring that the system has minimum latency to give a real- time experience and finding the best possible sound synthesis method.

This project aims to reduce environmental interference encountered in capacitive sensing, explore new areas for ultrasonic sensing in music industry and contribute to the research on gesture-based instruments.

- Contribute to **research on real-time gesture-based sound control**.

This project could inspire further research in touchless musical instruments and human-computer interaction (HCI) in music industry.

S/T methodology and associated work plan

Methodology

After the literature review, the project will follow a structured development process to ensure a well-designed, responsive, and low-latency interactive ultrasonic theremin.

1. **Initial Design & Planning**
    - Sketch a **complete initial design** on paper, mapping out the **working principle** and **playing experience** step by step.
    - Identify **necessary components** based on the desired functionality.
2. **Algorithm Development**
    - Define the system's logic in **pseudocode** to outline sensor data processing, pitch/volume mapping, and sound synthesis.
3. **Component Acquisition & Testing**
    - Purchase **all required components**, including alternative options for testing.

- Evaluate **ultrasonic sensor response times** for real-time accuracy.
- Compare **sound generation methods** (DAC, PWM, or software oscillators) to determine which provides **smoother, low-latency output**.

4. **Prototyping & Iteration**
   - Assemble the **initial system on a breadboard** and test the prototype.
   - Identify issues, refine the design, and retest until performance is satisfactory.
5. **PCB Design & Assembly**
   - Design a **custom PCB** to integrate all components into a single board.
   - Manufacture and assemble the PCB, ensuring all **connections and placements** are correct.
6. **Final System Integration & Testing**
   - Build the **physical structure** of the device, placing the **PCB, sensors, ports, and enclosure components** properly.
   - Conduct **final testing** of the complete system to ensure **functionality, stability, and playability**.

Work Package Descriptions

Detailed work package descriptions with contributors, objectives, task descriptions, etc.....

Work package number	**1**	Start date or starting event: 03.03. 2025		**Week 1**	
Work package title	**Literature Review**				
Participant number	1	2	3	4	5
Participant name	Aykut				
Weeks per participant	**1**				

Objectives • I will conduct further literature review on the device.
Description of work I will strengthen my knowledge on the subject by doing a literature review for 1 week. I think the most important point of this stage is to provide implementation ideas, algorithm options, etc. regarding the device's future processes.
Deliverables -
Milestones **M1.1**: I will be ready to start the initial design of the system.

Work package number	**2**	Start date or starting event: 10.03.2025		**Week 2**	
Work package title	**Initial design of the system**				
Participant number	1	2	3	4	5

Participant name	Aykut				
Weeks per participant	**1**				

Objectives

- Initial design

Description of work

T2.1 I will draw the initial design of the system going through the playing experience in my mind, thinking about the firmware algorithm and the circuitry to determine how the system look and work. After this, I will determine all the components I will need to implement my idea.

.

.

Deliverables

D2.1 A draft of the design and a possible BOM.

.

.

.

Milestones

M2.1 Components will be acquired and there will be a solid draft of the system to be built.

.

.

Work package number	**3**	Start date or starting event: 17.03.2025		**Week 3**	
Work package title	**Component Acquisition and Algorithm Creation**				
Participant number	1	2	3	4	5
Participant name	Aykut				
Weeks per participant	**1**				

Objectives

- While waiting for the components, I will write the algorithm I will use.

Description of work

T3.1 I will write the algorithm of the project as a pseudocode first then implement it with C so that I can start testing without wasting any time when my components arrive.

.

Deliverables

D3.1 The initial algorithm

.

.

.

Milestones

M3.1 Initial algorithm is created and components are acquired.

.

.

Work package number	**4**	Start date or starting event: 24.03.2025	**Week 4**

Work package title	**Testing out the components and sound generation methods**				
Participant number	1	2	3	4	5
Participant name	Aykut				
Weeks per participant	**2**				

Objectives

- Testing components and trying sound generation methods

Description of work

T4.1 The components will be tested one by one to be sure that they are all capable enough for the system to be built.

T4.2 I will test the wave generation methods such as DAC, PWM, Sound Synthesizer Chips, FM Synthesis, and analog oscillators.

Deliverables

D4.1 Sounds generated by different methods.

.
.
.

Milestones

M4.1 One of the most important parts of this project, the sound generation algorithm and method is determined.

.
.

Work package number	**5**	Start date or starting event: 7.04.2025		**Week 6**	
Work package title	**Initial Prototype**				
Participant number	1	2	3	4	5
Participant name	Aykut				
Weeks per participant	**4**				

Objectives

- Build an initial prototype as a proof of concept which shows that the final system will work as expected.

Description of work

T5.1 I will build the initial system on a basic breadboard which will generate sounds in different frequencies based on my hand movements and also will adjust the volume based on my other hand.

T5.2 I will start the design of the PCB while leaving room for final adjustments.

Deliverables

D5.1 Initial prototype

.
.
.

Milestones

M5.1 The hardest and most important step of the project will be completed. After this stage, everything will be debugging and design adjustments.

.
.

Work package number	**6**	Start date or starting event: 5.05.2025		**Week 10**	
Work package title	**System adjustments based on tests and feedback**				
Participant number	1	2	3	4	5
Participant name	Aykut				
Weeks per participant	**1**				

Objectives

- Final changes in the prototype will be made for the complete implementation

Description of work

T6.1 After making sure that the system works as expected in the previous month, I will make adjustments based on the given feedback and my test results.

T6.2 I am also planning to finish the PCB this week or next week latest.

Deliverables

D6.1 Final prototype version of the system without a PCB or a mechanical casing.
D6.2 Possibly the PCB.
.
.
.

Milestones

M6.1 The system is adjusted in accordance with the feedbacks and test results.
.
.

Work package number	**7**	Start date or starting event: 12.05.2025		**Week 11**	
Work package title	**Transition from prototype to designed device**				
Participant number	1	2	3	4	5
Participant name	Aykut				
Weeks per participant	**2**				

Objectives

- Transforming the prototype to a designed ready-to-use device.

Description of work

T7.1 I will use a basic casing for the circuitry in the shape of the initial design adjusted by feedbacks. It could be wood or a 3D PLA. The sensor and connector holes will be carved for wood and designed with them for PLA.

T7.2 The PCB version of the device will be tested, the PCB will already be designed and printed with necessary ports so that necessary algorithm or pin changes will not make it useless.

Deliverables

D7.1 After this work package, the device should be ready for use at least 4 days before the demo day.
.
.
.

Milestones

M7.1 Device is finished
.

.

Work package number	**8**	Start date or starting event: 02.06.2025		**Week 13**	
Work package title	**Project Report**				
Participant number	1	2	3	4	5
Participant name	Aykut				
Weeks per participant	**2**				

Objectives

- I will write the project report

Description of work

T8.1 I will write the project report using the images, data, videos, and everything I recorded and stored. In all weeks, I will be writing reports therefore this project report will be a completely detailed version of the whole building process using those reports and the device itself.

Deliverables

D8.1 Project Report

.
.
.

Milestones

M8.1 Ready to graduate

.
.

Demonstration

Performance Analysis Measures

1. **Responsiveness & Latency:** Measure sensor response time, processing delay, and overall system latency to ensure real-time playability.
2. **Sound Quality:** Test smooth pitch transitions and volume control precision.
3. **Accuracy & Playability:** Evaluate hand tracking precision, pitch range, and user experience.

Final Demonstration

- **Live Performance:** Show that the device works as expected by changing the pitch and volume and give them a chance to test it.
- **Technical Overview:** Explain system design, sensors, and sound synthesis.
- **Comparison & Results:** Explain the different approaches I could have taken, emphasizing why I chose this one for sensors, MCU, algorithm etc.

Impact

I believe the most important impact will be the inspiration for other gesture-based instruments projects. Other than that, methodologies tested for sound generation, low- latency etc. can also inspire other projects.

Risk analysis

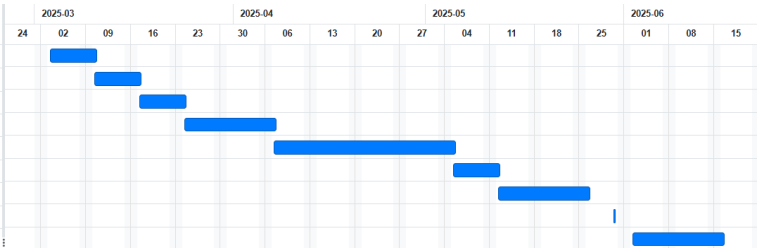
Most possible risk is sensor accuracy, especially for cheap ultrasonic sensors which have low accuracy: This will be handled using multiple sensors, filtering and data processing.

Pitch Jumps: This will be handled by the algorithm to smooth the frequency change process.

Sound Quality: Synthesis methods may have delay, this will be handled by using the most optimum method, than by optimizing the algorithm and also reducing the task load of the MCU.

Gantt Chart

ID	Task Name	Start	End	Duration
1	Literature Review	2025-03-03	2025-03-10	6 days
2	Initial design of the system	2025-03-10	2025-03-17	6 days
3	Component Acquisition and Algorithm Creation	2025-03-17	2025-03-24	6 days
4	Testing out the components and sound gener...	2025-03-24	2025-04-07	11 days
5	Initial Prototype	2025-04-07	2025-05-05	21 days
6	System adjustments based on tests and feedb...	2025-05-05	2025-05-12	6 days
7	Transition from prototype to designed device	2025-05-12	2025-05-26	11 days
8	Senior Design Day	2025-05-30	2025-05-30	1 day
9	Project Report	2025-06-02	2025-06-16	11 days



Ethical Issues

I do not think any aspect of this project will encounter any ethical issues in any engineering or non-engineering-related area.

References

[1] Lemelson-MIT Program, *"Leon Theremin"*, ***Lemelson-MIT*,** **https://lemelson.mit.edu/resources/leon-theremin,(accessed Mar. 4, 2025).**

[2]Moog Music Inc., *Etherwave Theremin User Manual*, **Asheville, NC, USA: Moog Music, 2020.**